

ASME PCC-2 · ON-STREAM PRESSURE-BOUNDARY REPAIR

Through-Wall Leak Repair of a 12" Cooling-Water Welding-Neck Flange

Engineering Case Study — Mechanical Piping Design & Analysis

1 SUMMARY

Project Classification Code	MC-REP-U46-CWS-032 — Line 12"-CWS-46-032-B1, Unit U46
Industrial Sector	Oil & Gas — Petroleum Refining (Cooling-Water circuit)
Geographic Location	La Pampilla Refinery, Ventanilla — Callao, Peru
Primary Technical Achievement	Design of an ASME PCC-2 (Art. 212) full-encirclement reinforcement collar that restored pressure-boundary integrity to a through-wall-leaking flange neck — repaired allowable pressure of 16.7 kg/cm² (1.63 MPa) , a ≈3.3× margin over operating pressure, executed with an on-stream, backing-free weld procedure.



FIG 1 As-found condition. 12" WN flange assembly on the cooling-water riser, Unit U46.



FIG 2 Defect. Localized wall loss and ≈3 mm through-wall perforation at the flange neck.

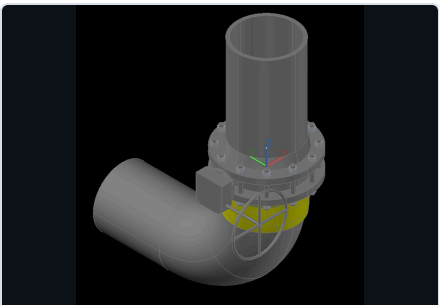


FIG 3 Engineered solution. 3D model of the two-piece reinforcement collar over the repair zone.

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TECHNICAL PARAMETERS & SPECIFICATIONS

PARAMETER	SPECIFICATION
Component & Rating	12" Welding-Neck flange, Class 150 RF — ASME B16.5; elbow/riser per ASME B36.10M
Material Composition	Riser pipe ASTM A106 Gr. B · Flange ASTM A105 · Reinforcement plate ASTM A516 Gr. 70, 8 mm
Service Fluid	Cooling water (CWS) — carbon-steel line, above-ground, uninsulated
Design Pressure Gradient	Operating 5 kg/cm ² (0.49 MPa); assessed up to flange rating 20 kg/cm ² (≈285 psi / 1.96 MPa)
Operational Temperature	Operating ≈25 °C; envelope ≤ 120 °C
Regulatory Compliance	ASME PCC-2-2022 (Art. 212, 206, 207, 210, 501) · ASME B31.3 · ASME B16.5 · ASME B36.10M · Repsol ED-B-06.00 / PE-B-0600.01
Technological Implementation	3D CAD model · parametric ASME PCC calculation engine (Excel) · DXF fabrication drawing

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PROJECT SCOPE

Problem Statement & Technical Constraints

A pressurized through-wall leak developed at the neck of a 12" welding-neck flange on cooling-water line 12"-CWS-46-032-B1 (Unit U46). Localized external corrosion had thinned the neck to perforation (≈3 mm defect), breaching the pressure boundary. The constraint set was demanding: the circuit had to **remain in service**, the geometry — a flange-to-elbow transition — ruled out a simple sleeve, and any repair had to be defensible by calculation against the code and the client's own engineering specifications before a welder ever struck an arc.



FIG 4 Corrosion morphology. External metal loss concentrated at the neck-to-flange fillet — the perforation origin.

MT-F02-2026/046/949

Unidad	N° de línea o equipo
U46 - Agua Enfriamiento	12"-CWS-46-032-B1
Fluido	Presión de operación (kg/cm2)
Agua de Enfriamiento	5,000

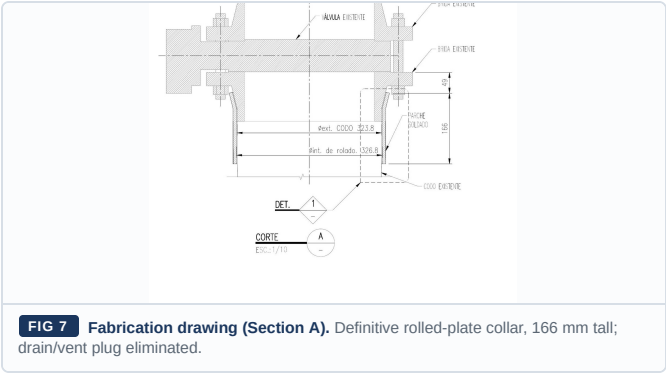
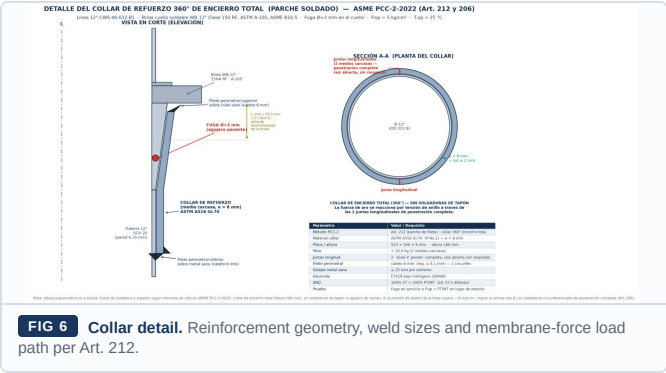
Equipo para reparación definitiva)

ombas 46P31A/B/C de agua de enfriamiento aguas arriba de válvula tipo mariposa. (10721256).

FIG 5 Process data sheet. Declared operating conditions (P = 5 kg/cm², T = 25 °C) governing the assessment.

Engineering Implementation & Strategic Solution

The repair was engineered as a **full-encirclement reinforcement collar** (ASME PCC-2, Article 212), fabricated from two ASTM A516 Gr. 70 half-shells (plate 523 × 166 × 8 mm each, ≈10.9 kg total) rolled to the riser radius and closed by two longitudinal CJP butt joints and two circumferential fillet welds. The collar was sized on a **membrane-force basis**: the reinforcement carries the pressure thrust released by the defect, $F = k \cdot P \cdot D_m = 75.85 \text{ N/mm}$, from which the minimum fillet leg (4.11 mm, specified at 6 mm) and axial overlap ($L_{\min} = 63.5 \text{ mm}$ each side) follow. Cold-forming strain of the rolled plate is 2.4 % — below the 5 % threshold, so no post-forming heat treatment is triggered. The weld procedure was deliberately specified **backing-free**: an open-root single-V groove, GTAW root (ER70S-6) and SMAW fill/cap (E7018), to keep the joint inspectable and avoid a crevice at the root.



16.7 kg/cm²
REPAIRED MAWP

3.3×
MARGIN VS. OPERATING P

4.11 mm
MIN. FILLET LEG

2.4 %
COLD-FORM STRAIN

APTO
FITNESS VERDICT

ENGINEERING INSIGHT

A reinforcement collar does not “patch a hole” — it **re-routes the pressure thrust** released by the defect into a welded shell, so the governing number is the membrane force $F = k \cdot P \cdot D_m$, not the size of the leak.

Every derived quantity was reproduced in a **parametric ASME PCC calculation engine** (Excel) with formula cells locked and each result annotated to its governing article — converting a one-off memo into a reusable, auditable standard for future collar repairs across piping and pressure vessels.

4 DELIVERABLES & TECHNICAL OUTPUTS

- ✓ **Three-Dimensional Modeling.** 3D CAD model of the elbow / WN-flange assembly with the reinforcement collar — isometric and sectioned digital twin used to fix geometry and clearances.
- ✓ **Analytical Reports.** 17-page design calculation memo (memoria de cálculo) to ASME PCC-2, plus a parametric ASME PCC calculation engine (Excel) with locked formulas and clause-referenced cells.
- ✓ **Technical Documentation.** Fabrication drawing (DXF + PDF), collar detail and weld-bevel details, and the technical specification (ET) for fabrication, welding and coating.
- ✓ **Project Administration.** Bill of Materials (BOM) for the two-piece collar and coating specification aligned to the Repsol painting standard (Scheme N°1, 260 μm DFT).

5 DOCUMENTATION & SUPPLEMENTARY DATA

- 1 **Dimensional Representation.** WN-flange dimensional specification (ASME B16.5, Class 150) and 3D isometric of the repaired assembly — basis for collar fit-up.
- 2 **Reinforcement / Stress Analysis.** Membrane-force sizing of the collar (F, minimum fillet leg, overlap length, maximum allowable pressure) per ASME PCC-2 Article 212.
- 3 **Site Implementation / Digital Rendering.** Field installation context and high-fidelity 3D section render of the collar seated over the repair zone.

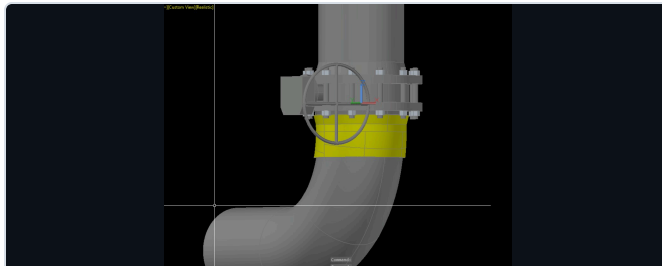


FIG 8 Digital rendering. Sectioned 3D view — collar band positioned across the flange-neck repair zone.

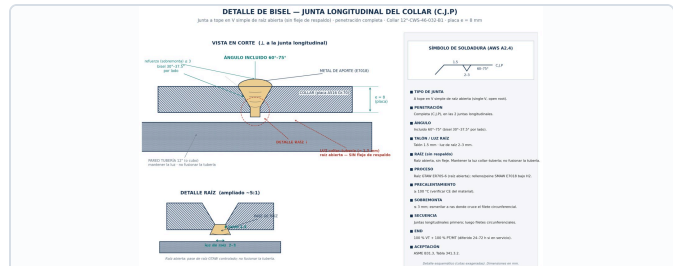


FIG 9 Weld-joint detail. Open-root single-V longitudinal joint — backing-free, GTAW root + SMAW fill.

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